Spectral Simulation Report

Automated Analysis

June 18, 2025

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1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot displayed in a full horizontal page and a detailed table summarizing the initial composition, actual positions, total pixels, and signal-to-noise ratio (SNR), along with simulation parameters such as temperature and electron density.

2 Sample: 8427

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	14798 K
Electron Density (n_e)	3.94e+17 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	35.85

Table 1: Simulation parameters and additional metrics for sample 8427.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Positions**
Al	0.00	0.0
Ar	0.00	0.0
C	2.58	848.0
Ca	9.44	150.0
Cl	3.60	35.0
Co	8.85	1404.0
Cr	6.09	397.0
Fe	10.57	2790.0
Mg	7.42	358.0
Mn	12.87	706.0
N	13.65	861.0
Na	1.92	325.0
Ni	1.72	478.0
O	0.00	0.0
S	9.73	228.0
Si	11.57	223.0
Ti	0.00	0.0
background	N/A	471.0
Total Positions		9274.0

Table 2: Composition analysis for sample 8427.

2.3 Spectral Plot

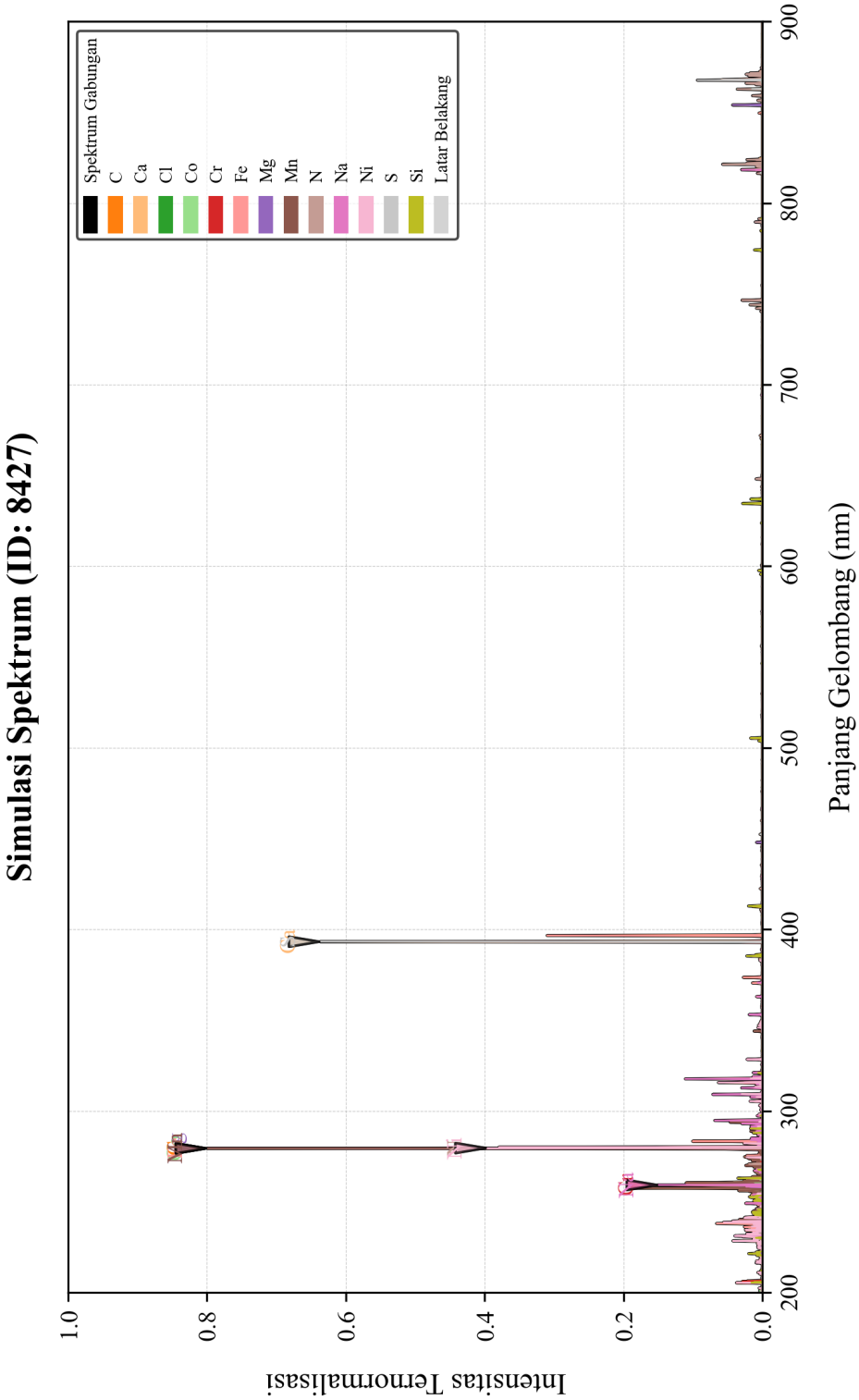

Figure 1: Spectral simulation plot for sample 8427 in full horizontal layout.

3 Sample: 5347

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	11869 K
Electron Density (n_e)	5.99e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	30.12

Table 3: Simulation parameters and additional metrics for sample 5347.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	10.52	462.0
Ar	8.28	727.0
C	3.88	447.0
Ca	0.35	150.0
Cl	10.96	35.0
Co	7.63	1404.0
Cr	11.04	397.0
Fe	10.10	2765.0
Mg	2.96	352.0
Mn	1.94	706.0
N	8.19	547.0
Na	1.34	322.0
Ni	2.04	476.0
O	4.83	217.0
S	7.25	228.0
Si	4.53	223.0
Ti	4.16	1186.0
background	N/A	371.0
Total Positions		11015.0

Table 4: Composition analysis for sample 5347.

3.3 Spectral Plot

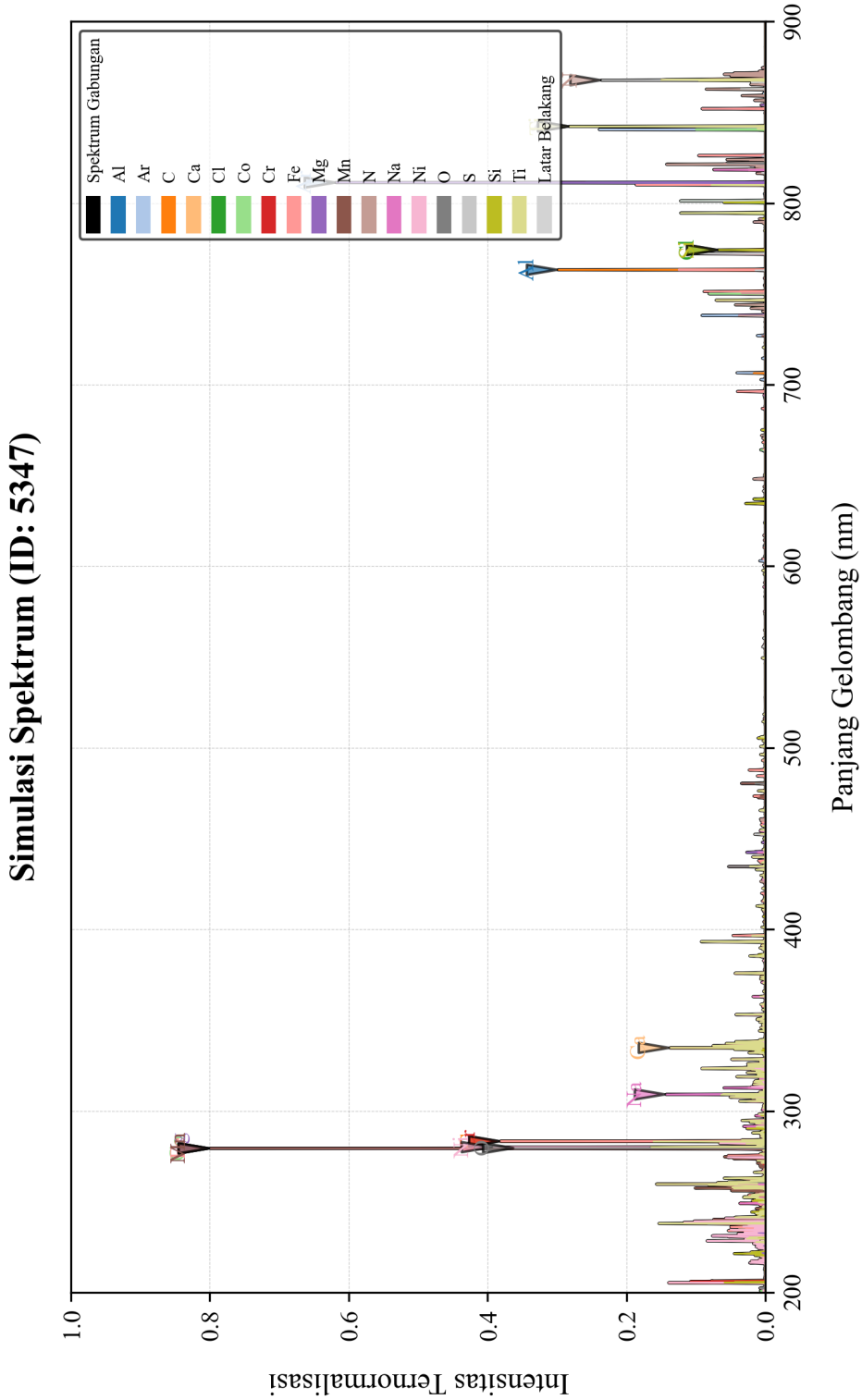

Figure 2: Spectral simulation plot for sample 5347 in full horizontal layout.

4 Sample: 8671

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	8232 K
Electron Density (n_e)	2.01e+16 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	36.83

Table 5: Simulation parameters and additional metrics for sample 8671.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	4.51	341.0
Ar	9.94	725.0
C	5.46	224.0
Ca	10.78	150.0
Cl	3.35	12.0
Co	10.37	1404.0
Cr	8.39	397.0
Fe	4.28	2725.0
Mg	0.00	0.0
Mn	8.14	706.0
N	2.70	396.0
Na	4.87	326.0
Ni	5.27	478.0
O	9.50	130.0
S	1.94	39.0
Si	7.96	220.0
Ti	2.55	1183.0
background	N/A	463.0
Total Positions		9919.0

Table 6: Composition analysis for sample 8671.

4.3 Spectral Plot

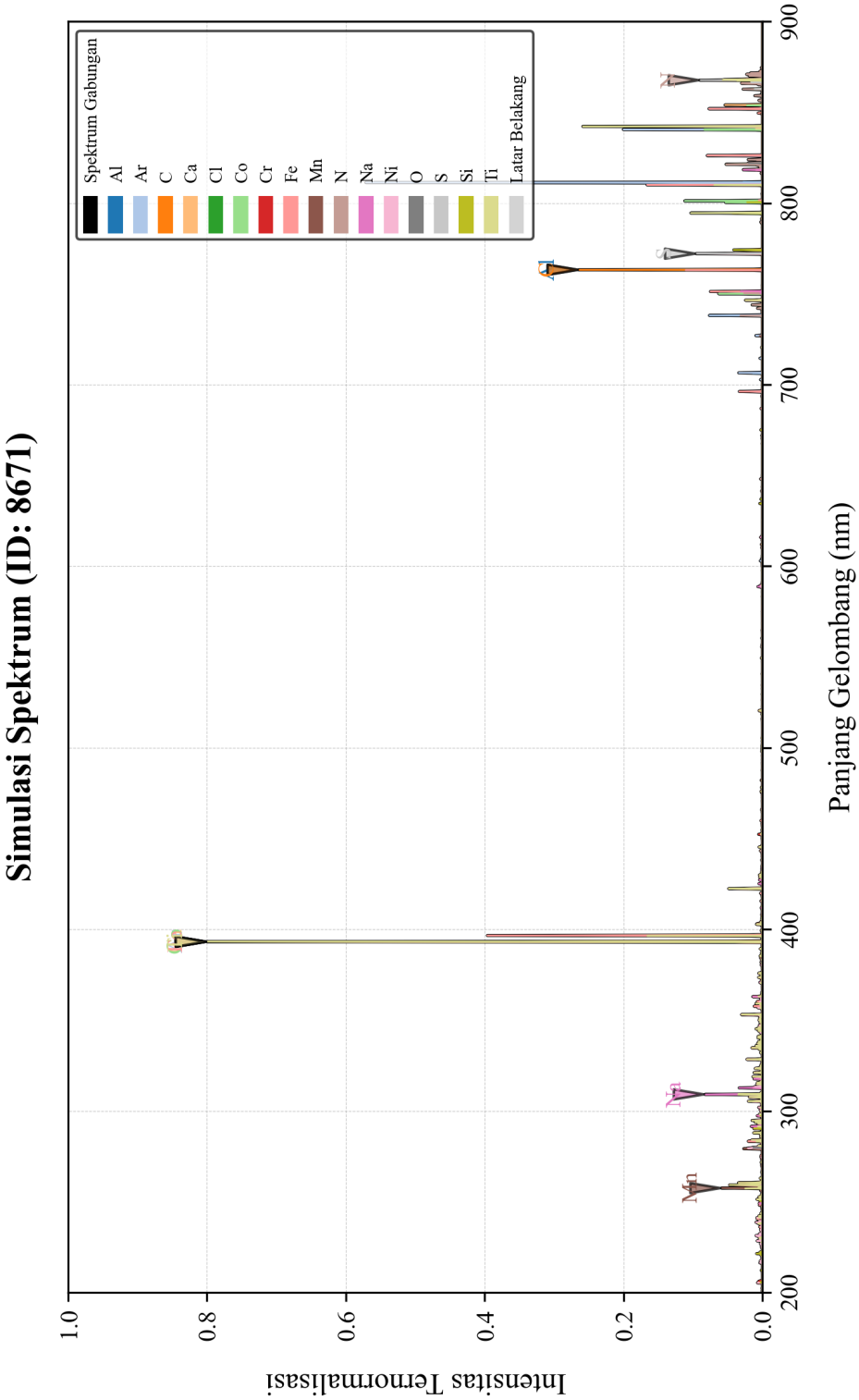

Figure 3: Spectral simulation plot for sample 8671 in full horizontal layout.

5 Sample: 7682

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	6414 K
Electron Density (n_e)	2.26e+17 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	35.34

Table 7: Simulation parameters and additional metrics for sample 7682.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	9.96	149.0
Ar	7.60	255.0
C	4.24	188.0
Ca	7.17	150.0
Cl	1.74	12.0
Co	3.86	828.0
Cr	6.15	397.0
Fe	3.71	2122.0
Mg	0.85	146.0
Mn	7.49	621.0
N	6.36	396.0
Na	4.80	330.0
Ni	10.64	478.0
O	2.01	12.0
S	4.43	25.0
Si	6.91	156.0
Ti	12.07	1181.0
background	N/A	926.0
Total Positions		8372.0

Table 8: Composition analysis for sample 7682.

5.3 Spectral Plot

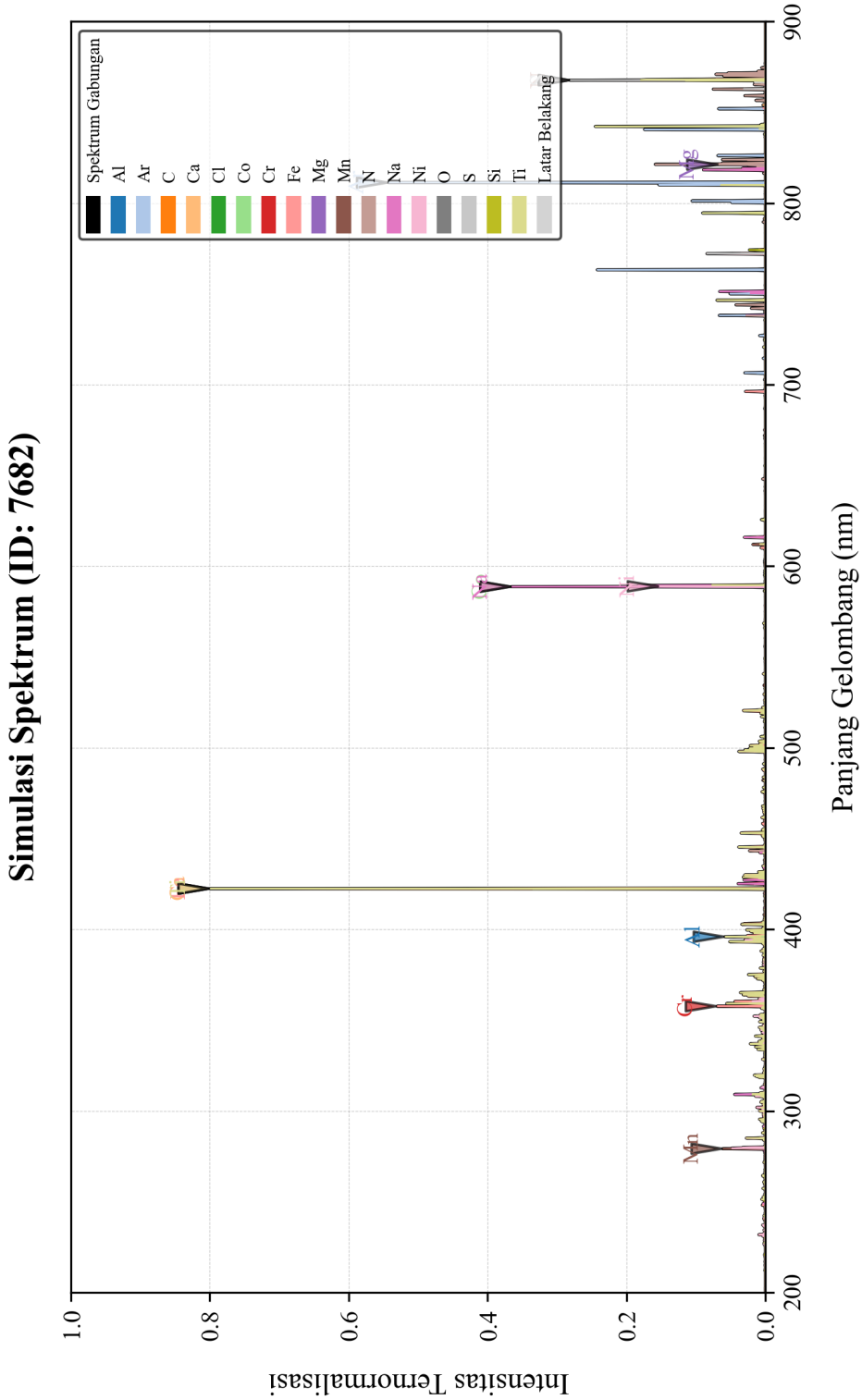

Figure 4: Spectral simulation plot for sample 7682 in full horizontal layout.

6 Sample: 18066

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	5101 K
Electron Density (n_e)	1.00e+18 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	43.58

Table 9: Simulation parameters and additional metrics for sample 18066.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.00	0.0
Ar	0.00	0.0
C	0.00	0.0
Ca	20.30	145.0
Cl	12.50	12.0
Co	1.44	623.0
Cr	21.64	397.0
Fe	0.00	0.0
Mg	2.48	125.0
Mn	11.99	359.0
N	0.00	0.0
Na	9.65	323.0
Ni	17.99	466.0
O	0.00	0.0
S	1.42	25.0
Si	0.58	83.0
Ti	0.00	0.0
background	N/A	2341.0
Total Positions		4899.0

Table 10: Composition analysis for sample 18066.

6.3 Spectral Plot

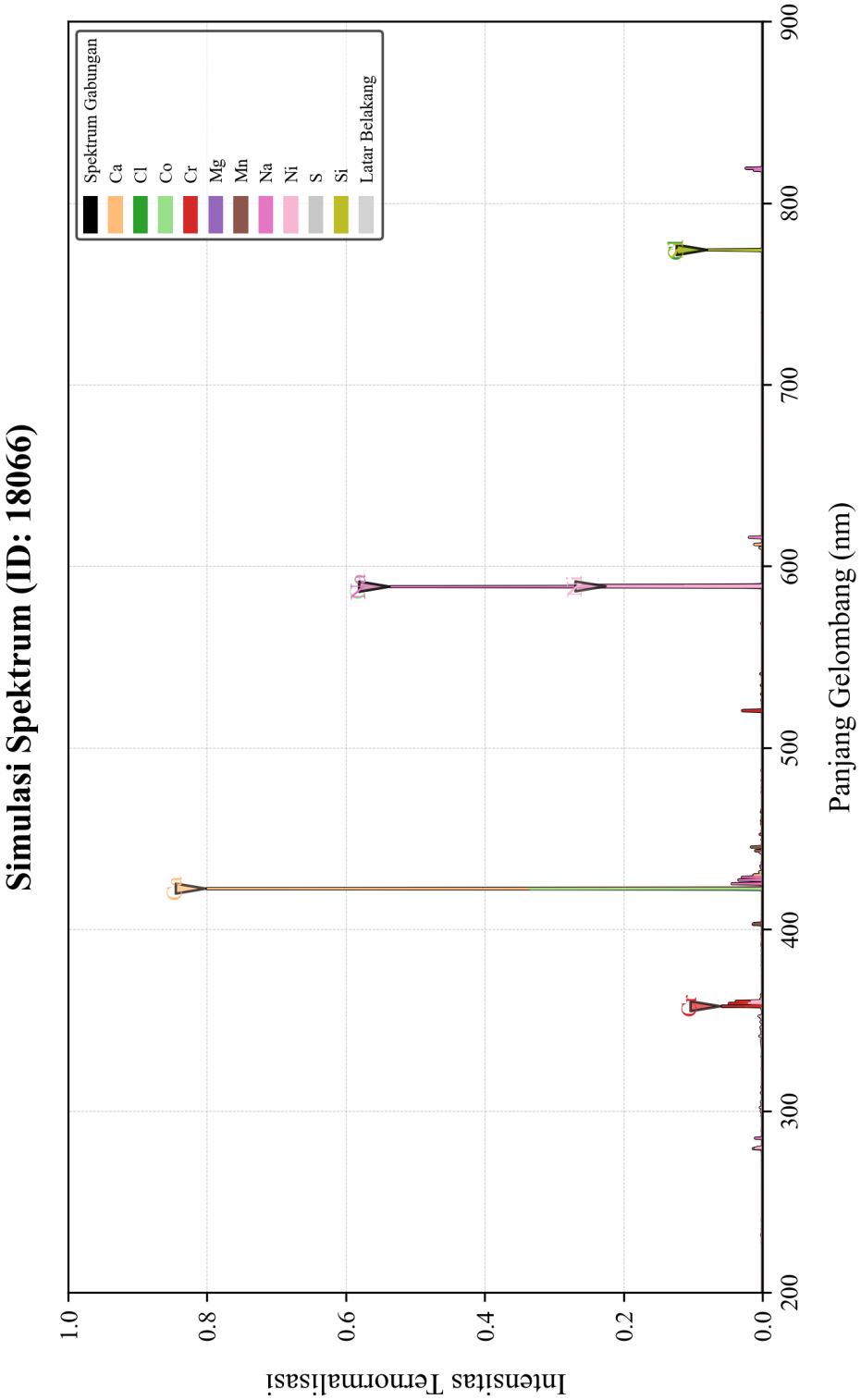

Figure 5: Spectral simulation plot for sample 18066 in full horizontal layout.